Today's agenda

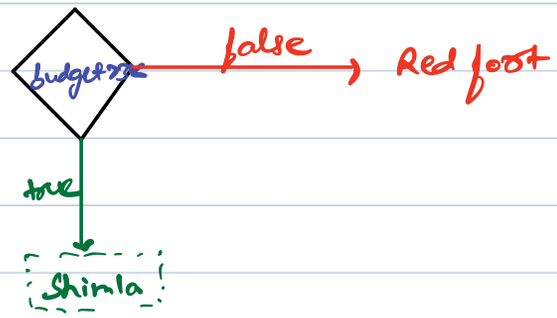
- ↳ if/else
- ↳ while loop
- ↳ for loop

AlgoPrep

// If - else

↳ budget > 3K → Shimla

budget < 3K → Red fort



↳ if we want to do something when cond^n is true and something else if Cond^n is false → if/else.

```
if (Cond^n) {
    //Statement 1
}
else {
    //Statement 2
}
```

if Cond^n : : true : Statement 1

if Cond^n : : false : Statement 2

AlgoPrep

Quiz 1:

true

```
if (15 > 7){
    s.o.p ("if");
}
else {
    s.o.p ("else");
}
```

→ if

Quiz 2:

```
int x = 70;
if (x > 70) {
    System.out.println ("if");
}
else {
    System.out.println ("else");
}
```

false

→ else

AlgoPrep

Q) Read a number and check if number is even or odd?

even no: divisible by 2 → remainder is 0
↳ n%2==0

odd no: not divisible by 2

```
Scanner scn = new Scanner(System.in);
int n = scn.nextInt();

if (n%2 == 0) {
    System.out.println("even");
}
else {
    System.out.println("odd");
}
```

AlgoPrep

// logical operator → && (and), || (or)

and operator (&&)

&& is false dominated

Cond^n 1 &&	Cond^n 2	ans
T	T	T
F	T	F
T	F	F
F	F	F

→ 3 Condn

Condn1 &&	condn2 &&	Condn3	answer
T	T	F	F
F	F	T	F
T	T	T	T

```
int x = 50;
int y = 30;
if ( x>60 && y>20){
    s.o.p ("if");
}
else {
    s.o.p ("else");
}
```

false

→ if

OR operator (||)

|| is true dominated

Cond^n 1	||	Cond^n 2	any
T		T	T
F		T	T
T		F	T
F		F	F

→ 3 Condn

Condn1	||	Condn2	||	Condn3	answer
T		T		F	T
F		F		T	T
T		T		T	T

AlgoPrep

Q) Read a number. If number is divisible by 2 or 3 Print "divisible", otherwise Print "not divisible".

```
Scanner scn = new Scanner(System.in);
    int n = scn.nextInt();

    if (n%2 == 0 || n%3==0) {
        System.out.println("divisible");
    }
    else {
        System.out.println("Not divisible");
    }
```

n = 4 -> divisible

n = 9 -> divisible

AlgoPrep

maths

marks : ↙

↳ marks > 90 → Exc.

↳ marks > 80, <= 90 → good

↳ marks > 70, <= 80 → avg.

↳ mark > 60, <= 50 → below Avg

if / Else if () .... / Else

```
if (Condn1) {
    //Statement 1
}
else if (Condn2) {
    // Statement 2
}
else if (Condn3) {
    // Statement 3
}
else {
    //Statement 4
}
```

Condn1 → true → exc. Statement 1 & skip remaining if else.
Condn1 → false → Check next condn

Condn2 → true → exc. Statement 2 & skip remaining if else.
Condn2 → false → Check next condn

Condn3 → true → exc. Statement 3 & skip remaining if else.
Condn3 → false → execute else

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Comparison

```
if (condn1) {        ← true
    //Statement 1
}
if (condn2) {        ← false
    //Statement 2
}
if (condn3) {        ← true
    //Statement 3
}
if (condn4) {        ← false
    //Statement 4
}
```

```
if (condn1) {        ← true
    //Statement 1
}
else if (condn2) {
    //Statement 2
}
else if (condn3) {
    //Statement 3
}
else if (condn4) {
    //Statement 4
}
```

↳ Condn1, Condn3 are true

↓

Statement 1

Statement 3

↓

Statement 1

AlgoPrep

Q) Max of 3 numbers

↳ Given 3 numbers, Print max out of these.

Ex:

a	b	c	ans
6	3	4	6
8	3	10	10
12	4	4	12
13	10	13	13
8	6	8	8
11	11	11	11

//Pseudo Code

```
Scanner scn = new Scanner(System.in);
int a = scn.nextInt();
int b = scn.nextInt();
int c = scn.nextInt();

if (a>=b && a>=c){
    s.o.p(a);
}
else if (b>=a && b>=c){
    s.o.p(b);
}
else {
    s.o.p(c);
}
```

a b c

~~13 10 13~~

→ a >= b && a >= c

a wins

→ b >= a && b >= c

b wins

→ c wins

Break till 9:25 AM

//Intro to loops

↳ Print all the numbers from 1 to 5.

```
System.out.println(1);
System.out.println(2);
System.out.println(3);
System.out.println(4);
System.out.println(5);
```

↳ Print numbers from 1 to 10^5.

loop: Do similar thing multiple times.

↳ while loop

↳ for loop

↳ ~~do while~~

Q) Print 1 to 5 using while loop.

loop controlling variable

```
int i = 1;
while (i<=5) {
    System.out.println(i);
    i = i+1;
}
```

```
int i = 1;
while (i<=5) {
    System.out.println(i);
    i = i+1;
}
```

↳ i = i+1 & i++ are same

i	i<=5	Print
1	T	1
2	T	2
3	T	3
4	T	4
5	T	5
6	F	
	↳ exit	

//Structure of while loop

1. initialize loop variable.
   int i = 1;

2. write while with condⁿ

```
while (i<=5) {

}
```

3. The statement you want to run.
   System.out.println(i);

4. updation of loop variable.

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Quiz 3:

```
int i=5;
while (i<10){
    System.out.print(i);
        i=i*2;
}
```

i	i < 10	Print
5	T	5
10	F Exit	

Quiz 4:

```
int i=1;
while (i<5){
    System.out.print(i+" ");
        i=i+2;
}
```

i	i < 5	Print
1	T	1
3	T	3
5	F Exit	

Quiz 5:

```
int i=1
while (i>=5){
    System.out.print(i);
        i++;
}
```

i	i >= 5	Print
1	F Exit	

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Quiz 6:

```
int i=1;
while (i<=5){
    System.out.print(i);
}
```

i	i<=5	Print
1	T	1
1	T	1
1	T	1
⋮	⋮	⋮

Quiz 7:

```
int i=0;
while (i<=5){
    System.out.print("AlgoPrep");
    i=i+1;
}
```

↳ How many times AlgoPrep will be Printed?

i	i<=5	Print
0	T	✓
1	T	✓
2	T	✓
3	T	✓
4	T	✓
5	T	✓
6	F	

↳ exit

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